

ANIRUDDH KHERA

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WORK EXPERIENCE

Tech Lead at TikTok (Org: Global Payments), California

Feb 2023 – PRESENT

- Scaled invoice system growing throughput from ~150K/day → 24M/day while reducing processing latency by >98% (24+ hrs → <5 mins) under peak load.
- Re-architected legacy cron-based system into distributed service-based execution, built precomputed billing framework for scheduled invoicing, enabled horizontal scalability for 70%+ growth and future demand (>200%).
- Improved resilience & reduced vendor cost via isolation architecture, retry framework, and Avalara caching, ensuring SLA (<15 min) and saving ~\$150K+ annually.
- Built observability, duplicate invoice detection, and LLM + MCP-based anomaly detection system for near real-time tax & invoice correctness validation.
- Built tax sub-systems from scratch; Enabled global real-time & offline sales tax calculations using self-built rule engine & 3rd party vendors; **Saved operational cost ~\$500k p.a.** by rule-engine & caching vendor responses; By-product improved **p50 latency by ~70%**
- Conceptualized new product capabilities of the invoicing system; Re-architected the legacy invoice system for flexibility & extensibility
- Designed distributed invoice cronjob-executors handling burst IO load ~**400,000/hr** invoices generation, making it horizontally scalable
- Monetized tax & invoice services to generate revenue **approximately \$150,000 p.a.** for the Global Payments org
- Ideated and architected shadow system to replay LIVE traffic in NRT onto shadow services (with n+1 release changes), to detect and prevent fund losses and PO issues. This service is expected to easily save \$1 million p.a. lost due to any new bugs introduced
- Improved **Payment Success Rate (PSR) by 15%** by analyzing payout metrics & building Payment Instrument (PI) pre-validation batch job to validate users' PI & further notify users via business (upstream) to update PI before auto-withdrawal scheduled job run

Senior Software Engineer (SE 3) at PayPal, California

Feb 2021 – Feb 2023

- Identified hidden bottlenecks impacting p95 & p99 of APIs; Designed & implemented solution **improving overall latency by 22%-56%**
- Eliminated redundant ~**100 mill network hops/day** which were for loading user profiles during the API call to calculate fees
- Analyzed data distribution & data access patterns and proposed a solution to **improve the hit ratio of L1 cache by 15%-20%**
- **Led a team of 4 engineers** to implement the above caching strategy; Proposed the high-level design for the new caching strategy
- Built Rule Engine from scratch for Treasury to customize forex rates blend logic per currency level; Migration from hardcoded batch
- Enhanced the Rule Engine further to create derivative forex rates for each currency; Enabling market competitive rates for large merchants thereby generating an **additional \$300 million revenue p.a.**

Software Engineer 2 at PayPal, California

Feb 2019 – Feb 2021

- Developed externalized APIs to allow sellers list products on global e-comm platforms by providing quote rates (ForEx) with markup
- Enhanced x-currency transactional APIs with guaranteed forex rates, calculating revenue payouts with **latency 95%ile being 25ms**
- Reduced overall Pricing's Failed Customer Interactions (FCIs) **by +20%** by building generic and configurable retry-mechanism on the downstream calls, FCIs per high traffic APIs **reduced by ~85%**
- Brought down by ~**75%** (1 hr to 15 min) the total runtime of the regression test suite by parallelizing test runs and reducing I/Os

Software Engineer (Machine Learning) at 23andMe, California

Jun 2018 – Aug 2018

- Proposed solution to plugin arbitrary trained machine-learning models, agnostic of ML frameworks, languages, and serialization techniques, managing lifecycle of models and serving predictions with low latency via REST APIs; Zeroing developers' efforts

Senior Data Engineer at GE Healthcare, India

Oct 2016 – Dec 2016

- Streamlined the ETL processing of medical devices' log data by developing UDFs / modules in Java and batch processing the data on Hadoop using Sqoop and Hive to monitor devices' health

Software Developer at GE Healthcare, India

Jul 2014 – Oct 2016

- Implemented RESTful APIs using Spring Boot and built UI components using React & Redux framework
- Developed a push-notification system using RabbitMQ and Socket.io to track medical risks and patients' appointments
- Reduced technical debt of the codebase to ~10% using Sonar; Wrote Junit test cases using Mockito with 95% code coverage
- Conducted market research to understand oncology workflow and scope requirements from the top 10 healthcare providers
- **Led a team of 6 engineers** functioning as **Lead Software Engineer**; Recognized with **2 awards** for the impact on the project

TECHNICAL SKILLS

Programming Languages	: Java, Go, Python, C/C++, Scala
Web Tech & Tools	: REST web services, Thrift RPC, Git, Splunk, Juno (distributed cache), Hadoop, Spark, Kafka, RMQ
Frameworks	: Spring boot, Kite, Gin, Play, Flask, Maven, Hibernate, JPA
Databases & Cloud Tech	: MySQL, Oracle, PostgreSQL, MongoDB; AWS (EC2, ECS, Fargate, DynamoDB, S3), Docker

EDUCATION

Master of Science, Computer Science

New York University, Courant Institute of Mathematical Sciences, New York

Dec 2018

GPA: 3.74/4.0

ANIRUDDH KHERA

Bachelor of Engineering, Computer Science& Engineering
Manipal Institute of Technology, India

May 2014
GPA: 7.55/10

ACHIEVEMENTS & EXTRA-CURRICULAR

- Published research study paper on Bayesian Neural Network <https://arxiv.org/abs/1801.07710> Jan 2018
- **Won 3rd prize at HackNY Hackathon**; Built word embedding model predicting visa interview outcome using chatbot Apr 2018
- Graduated from *Edison Engineering Development Program*, one of the premier **leadership programs at GE**
- Achieved 2 silvers and 1 bronze medal in **National swimming meet, India**; Won 100+ medals in state & college events
- Appointed as a **Teaching Assistant** for graduate courses on **Predictive Analytics** and **Big Data** at NYU